

This document summarizes the additions and modifications I made to the utils files from Jannik Wirthheim's original implementation, which this thesis extends.

Dataset_utils.py

- Added pad_expert_tensors, which pads predator/prey expert tensors up to max_pre agents and inserts the "active" mask channel.
- Added pad_rollout_tensors, which is the same padding, but for tensors that already have the active channel.
- Removed process_frame and filter_frames (Jannik's YOLO/DeepSORT video-tracking functions). This file now starts from the hand-labeled/expert-tensor pipeline instead.

Encoder_utils.py

- TrajectoryAugmentation: now uses 6-feature prey input with an “active” mask channel (previously 5 features, no active channel).
- sample_data: now takes idx =None and returns (batch, indices) instead of just batch.
- train_encoder: batch size increased from 10 → 128.
- Added logic to protect “active” mask from noise/dropout augmentation.

Eval_utils.py

- sample_data calls updated to handle the new (batch, indices) return value.

Experiments_utils.py

- I created new experiments to evaluate learned predator-prey behavior. I removed most of Jannik's functions.
- Added geometry/response helpers: _as_float_tensor, _unit_vectors, headings_from_velocity, prepare_geometry_cache, compute_continuous_predator_response_tensor.
- Added pursuit/evasion metric computation: compute_closing_speed, compute_predator_nearest_distance, compute_pursuit_alignment, compute_pursuit_metrics, pursuit_metrics_from_simulation, safe_polarization, moving_average.
- Added hand-labeled attack-clip analysis: load_attack_frames, compute_attack_expert_metrics.
- Added individual metric plots (one model-comparison plot per metric, replacing the old combined plotters): plot_angular_momentum, plot_closing_speed, plot_degree_of_sparsity, plot_distance_to_predator, plot_escape_alignment, plot_nearest_pre_distance, plot_polarization, plot_pre_reorientation, plot_pursuit_alignment, plot_metric_boxplot.
- Changed logic of compute_pin_an_maps and plot_policy_maps.

Train_utils.py

- alive_mask: new helper to ignore padded agents when computing rewards.
- gradient_estimate: added theta_norm_ref argument to use a fixed reference norm for clipping (improves stability).
- discriminator_reward:
 - Supports joint predator-prey discriminators.
 - Properly handles padding via active masks.
 - Cleaner separation between “reward-only” and “combined reward” modes.

- `optimize_es`:
 - Updated to handle padding and joint discriminators.
 - Added `n_pre = 32` parameter.
- `pretrain_policy`:
 - Now filters out padded agents before training.
- `calculate_metrics`:
 - Evaluates over multiple episodes (captures stochasticity). Added `n_episodes=5` parameters. Rollout generation is run in a loop over `n_episodes`.
 - Supports configurable `n_pre` and explicit padding. Added `n_pre=32`.
- `gradient_penalty_joint`: new function for joint discriminator gradient penalty.

Sim_utils.py

- `get_state_tensors`: updated to account for padding/active channel.
- `run_env_simulation`: updated to handle the active channel in trajectories.

Vec_sim_utils.py

- `get_state_tensors`: handles padding/active channel.
- `run_env_vectorized`: handles padding/active channel.
- `run_batch_env`: updated for active-channel tensors.
- `apply_perturbations`: added `n_pre=32` parameter.

Couzin_utils.py, Mmd_loss.py

- None.